

ATUL KUMAR

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EDUCATION

Birla Institute of Technology, Mesra

2023 – Present

Bachelor of Technology in Chemical Engineering

Ranchi, India

Chinmaya Vidyalaya, Bokaro Steel City

2021 – 2022

Higher Secondary: Class: 11,12

Bokaro, Jharkhand

EXPERIENCE

Web Development Intern

Apr 2026 – Jun 2026

Zaalima Development (Remote)

- Developed an AI-powered Chrome Extension Generator using React, TypeScript, and Tailwind CSS, automating the generation of 5+ core extension files from natural language prompts and reducing project setup time by 80%.
- Implemented a modular code generation engine supporting 20+ Chrome Extension APIs with reusable templates and validation, enabling rapid creation of production-ready browser extensions.
- Owned the design and delivery of an enterprise SOP knowledge assistant (RAG-based), from retrieval architecture through deployment, cutting manual SOP lookup time for internal teams.

PROJECTS

Stock-Research-Assistant | *Python, FastAPI, LLMs, Financial APIs* 🐙

Jul 2026

- Developed a Stock Research Assistant integrating 4+ analysis modules for technical and fundamental analysis.
- Built a FastAPI backend with 10+ REST API endpoints for automated stock data processing and AI-driven research reports.
- Implemented LLM-based summarization analyzing 8+ financial indicators, reducing manual research time by 70%.
- Designed a modular architecture integrating external APIs for real-time market data and explainable insights.

SOP-RAG-Assistant | *FastAPI, MongoDB, LLMs, Vector Embeddings* 🐙

May 2026

- Built an enterprise RAG assistant using FastAPI and LangChain with a 5-stage retrieval pipeline for SOP search.
- Implemented semantic search via vector embeddings, improving response relevance by 40% over keyword baseline.
- Built an end-to-end document pipeline (PDF parsing → chunking → embedding generation), reducing retrieval latency by 35%.
- Enforced source-grounded response generation via retrieval-constrained prompting, ensuring every answer traces to a cited SOP section.

Dental Caries Segmentation | *YOLOv8, PyTorch, Computer Vision* 🐙

Jan 2026

- Developed a YOLOv8-based segmentation pipeline for detecting dental caries from 750+ radiographic images.
- Trained on 768×768 images using AdamW optimizer with cosine learning-rate scheduling.
- Applied augmentation techniques to increase effective dataset diversity by approximately 4×.
- Achieved 22.6% mask recall using Precision, Recall, and mAP evaluation metrics.

TECHNICAL SKILLS

LLM & GenAI: LangChain, Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, Prompt Engineering, LLM Integration

Machine Learning & Computer Vision: PyTorch, Scikit-learn, YOLOv8, Computer Vision, Object Detection, Image Segmentation, Natural Language Processing

Programming Languages: Python, C++, C, JavaScript, HTML, CSS

Backend & APIs: FastAPI, REST APIs

Databases: MongoDB, MySQL

Tools: Git, GitHub, Jupyter Notebook

CERTIFICATIONS & JOB SIMULATIONS

Deloitte Data Analytics Job Simulation (Forage)

Data Analyst Bootcamp (Udemy)